

Script-Controlled ACL – Restrict Record Access Based on Field Value

1. Ideation Phase

1.1 Project Idea

The project provides field-value based record access control using a script-controlled Access Control List (ACL). Instead of giving every user the same access, the system checks selected record fields and decides whether a user can view, edit, or manage that record.

1.2 Problem Statement

Traditional role-based access control may give users access to an entire table or module. In some applications, access must depend on the value stored in each individual record, such as department, branch, owner, status, or confidentiality level.

1.3 Proposed Solution

A script evaluates the current user's identity, role, and selected field values. Based on the result, an ACL rule allows or denies the requested operation.

1.4 Objectives

- Restrict individual records according to field values.
- Support view, create, update, and delete permissions.
- Apply rules consistently through server-side validation.
- Reduce unauthorized access to sensitive records.
- Keep the access rules configurable and maintainable.

1.5 Example Use Case

If an employee record contains Department = HR, an HR user can access it, while a user from another department is denied. Similar rules can be applied using Branch, Owner_ID, Status, or Confidentiality_Level.